E-Vex

Solveing the follwing System:

$$\begin{cases} \log_2(x) + \log_2(y) = 2 \\ x + y = \frac{15}{2} \end{cases}$$

$$\log_2(x) - \log_2(y) = 2 \Longrightarrow \log_2(x \cdot y) = 2$$

then: $\dfrac{\ln(x \cdot y)}{\ln(2)} = 2$

$$\Longrightarrow \ln(x \cdot y) = 2\ln 2 \Longrightarrow \ln(x \cdot y) = \ln 4$$

then: $e^{\ln(x \cdot y)} = e^{\ln 4} \Longrightarrow x \cdot y = 4$

after:

$$\begin{cases} x \cdot y = 4 \\ x + y = \frac{15}{2} \end{cases} \Longrightarrow y = \frac{15}{2}(x)$$

$$x\left(\frac{15}{2} - x\right) = 4$$

$$\frac{15x}{2} - x^2 - 4 = 0 \xRightarrow{x \cdot -1} x^2 - \frac{15x}{2} + 4 = 0$$

$$2\left(x^2 - \frac{15x}{2} + 4\right) = 0$$

$$2x - 15x + 8 = 0$$

$x = \dfrac{15}{4}$ and $x = \dfrac{\sqrt{161}}{4}$

when $x = \frac{15}{4}$:

$$\frac{15}{4} + y = \frac{15}{2}$$

$$y = \frac{15}{2} - \frac{15}{4}$$

$$y = 3.75$$

when $x = \frac{\sqrt{161}}{4}$

$$\frac{\sqrt{161}}{4} + y = \frac{15}{2}$$

$$y = \frac{15}{2} - \frac{\sqrt{161}}{4}$$

$$y = 4.32$$

$$S: \left(\frac{15}{4}, 3.75\right) \left(\frac{\sqrt{161}}{4}, 4.32\right)$$